

Problem 1

```
public class Trace1 {

    public static void main(String[] args) {
        String text = "Bracu";
        System.out.println(change(text));
    }

    public static String change(String str) {
        char ch = str.charAt(1);
        String part = str.substring(2, 5);

        System.out.println(ch);
        System.out.println(part);

        return ch + part;
    }
}
```

Problem 2

```
public class Trace2 {

    public static void main(String[] args) {
        int[] arr = {10, 12, 14, 15, 17};
        String var1 = addToArray(arr, 25);
        System.out.println(var1);
    }

    public static String addToArray(int[] a, int num) {

        int[] new_arr = new int[a.length + 1];

        for (int i = 0; i < a.length; i++) {
            new_arr[i] = a[i];
        }

        new_arr[a.length] = num;

        System.out.println(new_arr[2]);
        System.out.println(new_arr[5]);

        return "The statement is True";
    }
}
```

Problem 3

```
public class MainClass {

    public static void main(String[] args) {
        int[] b = {20, 5, 15, 30, 13};
        System.out.println(metA(b, 0));
    }

    public static int metA(int[] a, int i) {

        if (i == a.length - 1) {
            return a[i];
        }

        int x = metA(a, i + 1);

        System.out.println(a[i] + x);

        if (a[i] < x) {
            return a[i];
        } else {
            return x;
        }
    }
}
```

Problem 4

```
public class Tracing4 {

    public static void main(String[] args) {
        int[] arr = {3, 4, 1, 2, 5};
        recur(arr, 0, 0);
    }

    public static void recur(int[] b, int i, int s) {
        if (i >= b.length) {
            return;
        }
        int val = b[i];
        b[i] = s;
        recur(b, i + 1, s + val);
        System.out.println(b[i]);
    }
}
```

Problem 5

```
public class MethodTracingA {

    public static void main(String[] args) {

        m1(1);

    }

    public static int m1(int n) {

        int b = 2;

        if (n <= 5) {

            b = m1(n + 1) + m2(n + b);

            System.out.println(b);

        } else {

            return b;

        }

        return b;

    }

    public static int m2(int a) {

        int b = a + a % 2 * 3;

        return a - b * a;

    }

}
```

Problem 6

```
public class TraceQA {

    public static void main(String[] args) {

        System.out.println(foo(4));

    }

    public static int foo(int n) {

        int sum = 0;

        if (n > 0) {

            int left = foo(n - 1);

            int right = bar(n + left);

            sum = left + right;

            System.out.println(sum);

        }

        return sum + n;

    }

    public static int bar(int x) {

        int y = x * 2;

        return x - x * y;

    }

}
```

Problem 7

```
public class Quiz8 {

    public static void main(String[] args) {

        System.out.println("Exam Starts");

        System.out.println(mA(11, 31));

        mB(3);

        System.out.println("Exam Ends");

    }

    public static int mA(int a, int b) {
```

```
        if (a > b) {

            return a - b;
        }

        else if (a % 2 == 1) {

            mB(a + 2);

            return a + mA(a + 3, b);
        }

        else {

            mB(b - 1);

            return b - mA(a, b - 4);
        }
    }

    public static void mB(int n) {

        System.out.println("mB: " + n);
    }
}
```

Problem 8

```
import java.util.Arrays;

public class Trace9 {

    public static int myst(String str, int[] arr, int idx) {

        if (idx == arr.length) {
            System.out.println("Reached");
            return 0;
        }
        char ch = str.charAt(idx);
        if (Character.isUpperCase(ch)) {

            arr[idx] = arr[idx] * 2;

        } else {

            arr[idx] = arr[idx] + 1;
        }
        System.out.println(
            "Forward: " + idx + " -> " + arr[idx]
        );

        int sum = myst(str, arr, idx + 1);

        arr[idx] = arr[idx] - 1;

        System.out.println(
            "Backward: " + idx + " -> " + arr[idx]
        );
        return arr[idx] + sum;
    }

    public static void main(String[] args) {
        int[] nums = {2, 3, 4, 5};

        String text = "AbCd";

        int result = myst(text, nums, 0);

        System.out.println(Arrays.toString(nums));
        System.out.println(result);
    }
}
```

Problem 9

```
public class Trace10 {

    public static void main(String[] args) {

        String text = "aBcD";

        int[] nums = {2, 3, 1, 4};

        int result = mystery(text, nums, 0);

        System.out.println("Result = " + result);

        System.out.println(Arrays.toString(nums));
    }

    public static int mystery(String str, int[] arr, int i) {

        if (i == arr.length) {

            System.out.println("Base");

            return 1;
        }

        int old = arr[i];

        if (Character.isUpperCase(str.charAt(i))) {

            arr[i] = arr[i] + i;

        }
    }
}
```

```

    } else {

        arr[i] = arr[i] * 2;
    }

    System.out.println(
        "F " + i + " " + arr[i]
    );

    int x = mystery(str, arr, i + 1);

    int y = calculate(str.charAt(i), arr[i]);

    arr[i] = old + x % 3;

    System.out.println(
        "B " + i + " " + arr[i] + " " + x + " " + y
    );

    return arr[i] + x + y;
}

public static int calculate(char ch, int n) {

    if (Character.isUpperCase(ch)) {

        return n - 1;
    }

    return n + 2;
}
}

```

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Editing in progress

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